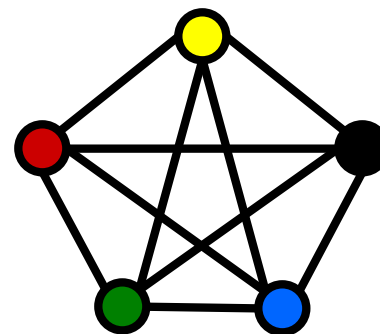
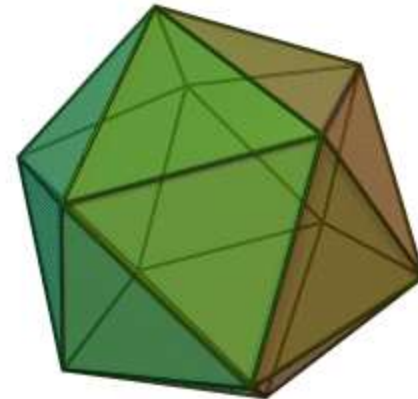
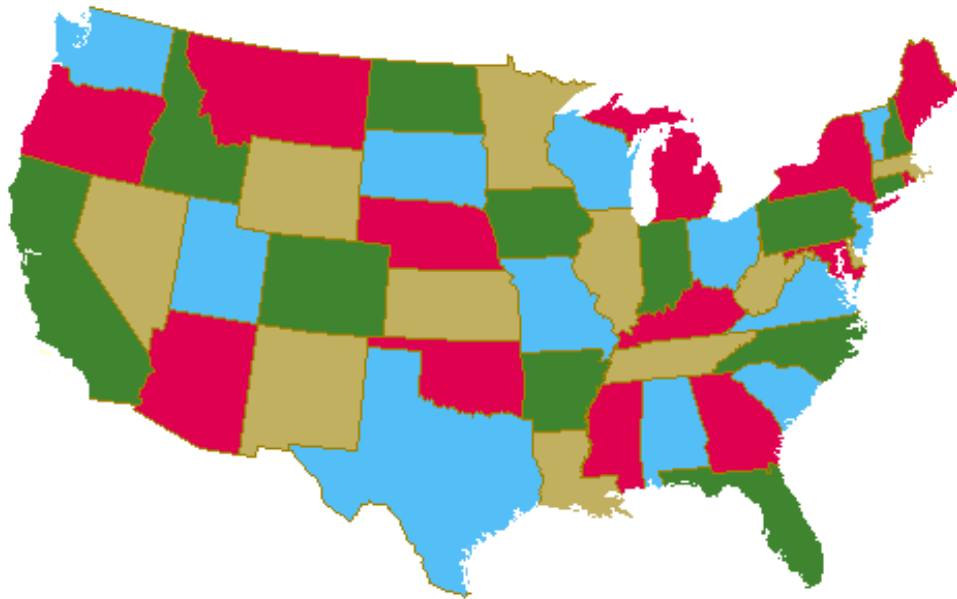


Graph Colouring

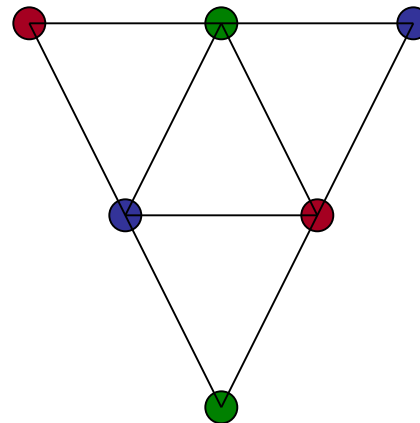
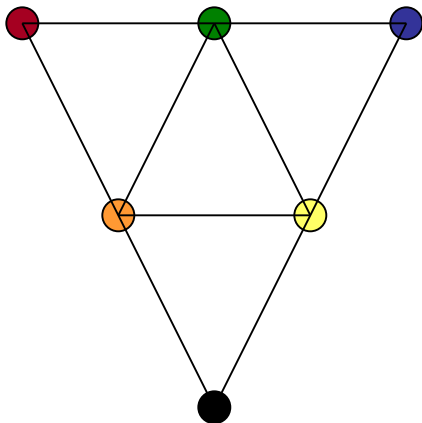


Graph Colouring

Graph Colouring Problem:

Given a graph, colour all the vertices so that two adjacent vertices get different colours.

Objective: use **minimum** number of colours.

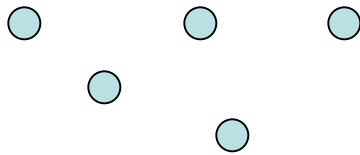


3-colourable

Optimal Colouring

Definition. min #colors for G is chromatic number, $\chi(G)$

What graphs have chromatic number one?



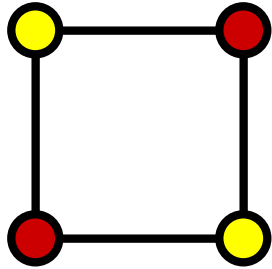
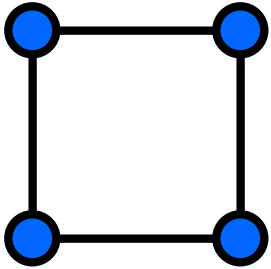
when there are no edges...

What graphs have chromatic number 2?

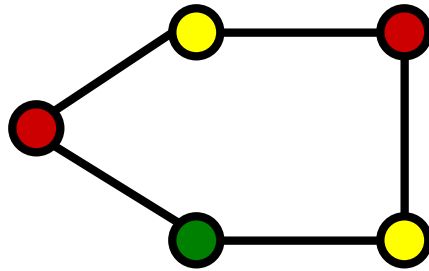
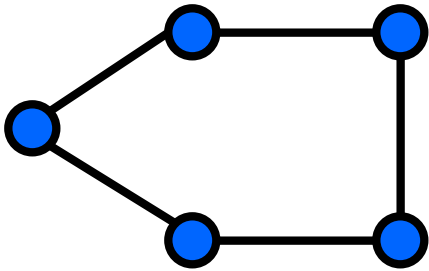
What graphs have chromatic number larger than 2?

A path? A cycle? A triangle?

Simple Cycles

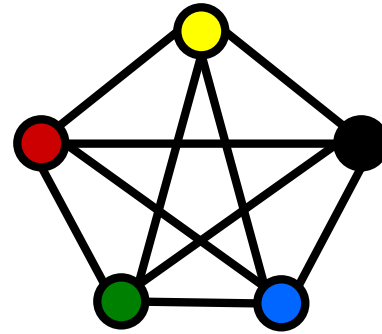
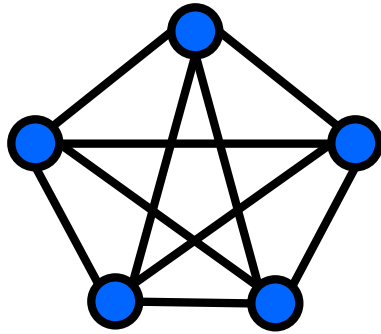


$$\chi(C_{\text{even}}) = 2$$



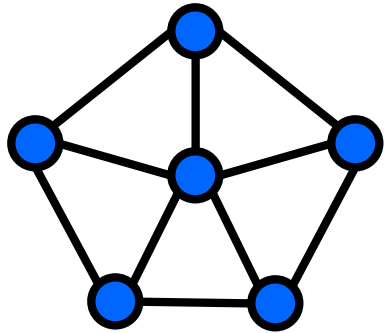
$$\chi(C_{\text{odd}}) = 3$$

Complete Graphs

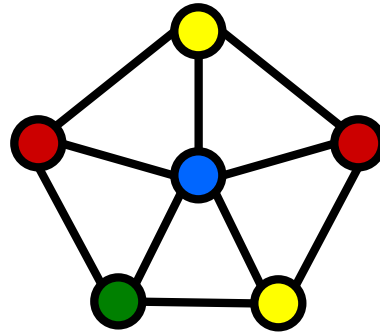


$$\chi(K_n) = n$$

Wheels

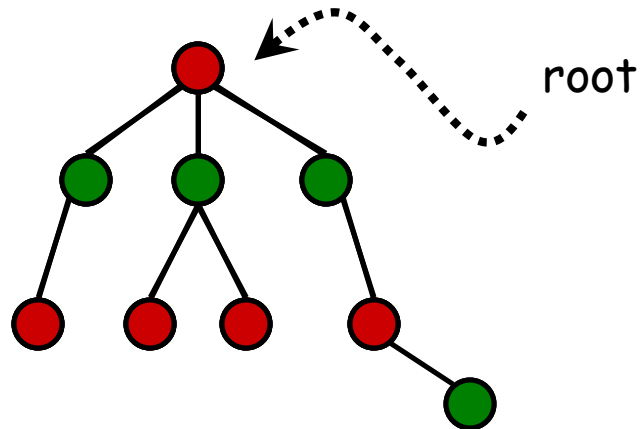


W_5



$$\chi(W_{\text{odd}}) = 4 \quad \chi(W_{\text{even}}) = 3$$

Trees



Pick any vertex as "root."

if (unique) path from root is

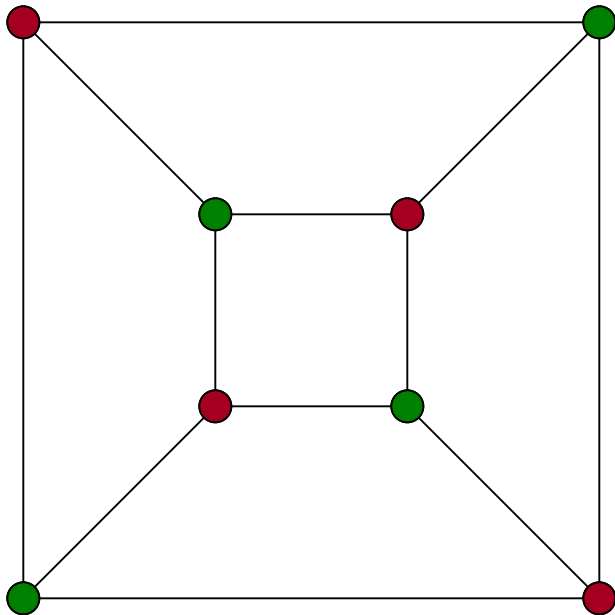
even length: ●

odd length: ●

Can prove more formally using induction.

2-Colourable Graphs

When exactly is a graph 2-colourable?



This is 2-colourable.

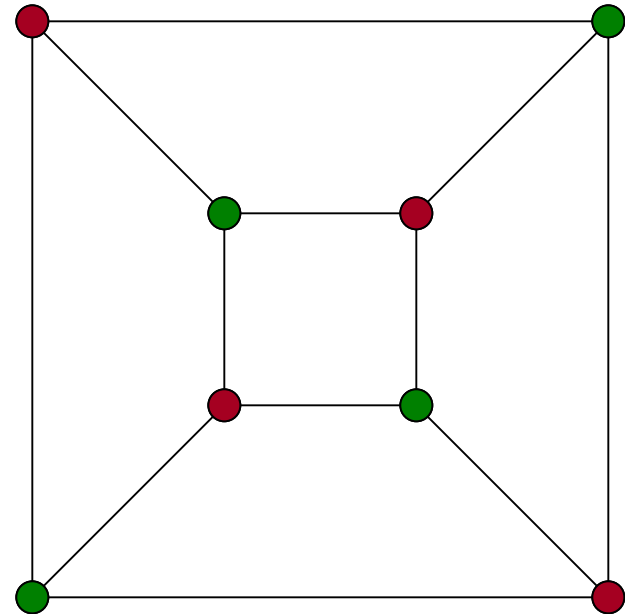
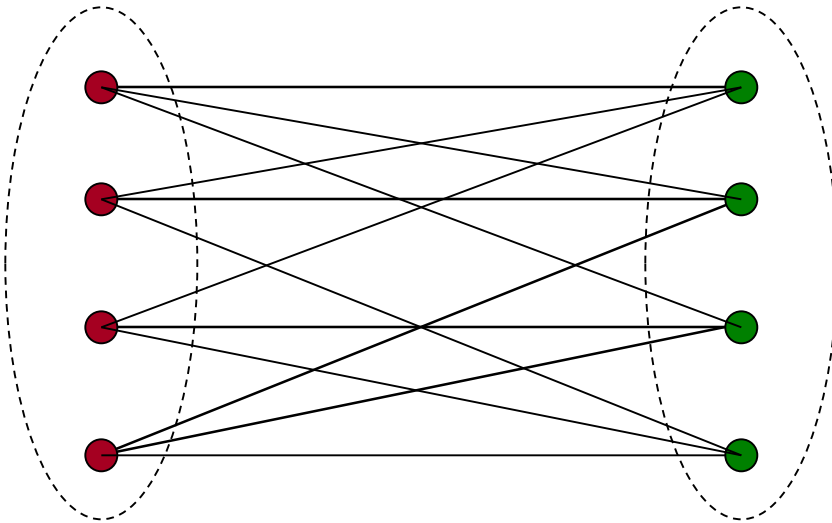
2 colourable: tree, even cycle, etc.

Not 2 colourable: triangle, odd cycle, etc.

Bipartite Graphs

When exactly is a graph 2-colourable?

Is a bipartite graph 2-colourable?

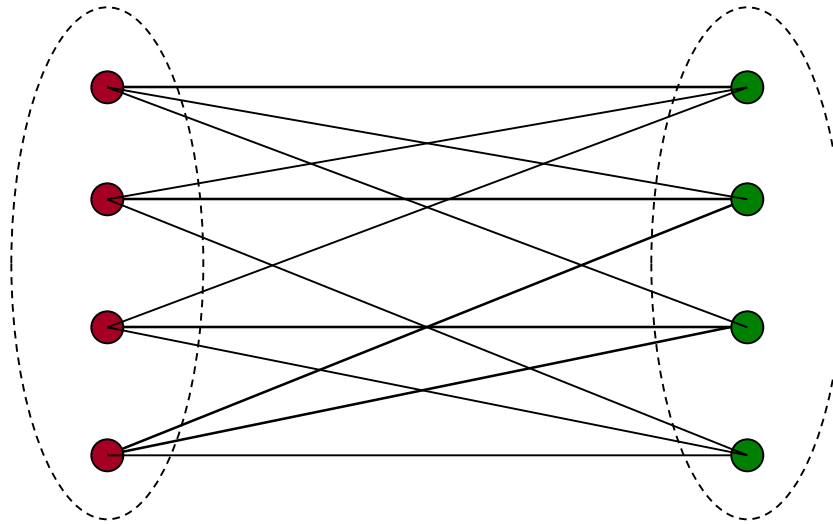


Is a 2-colourable graph bipartite?

Fact. A graph is 2-colourable if and only if it is bipartite.

Bipartite Graphs

When exactly is a graph **bipartite**?



Can a bipartite graph has an odd cycle?

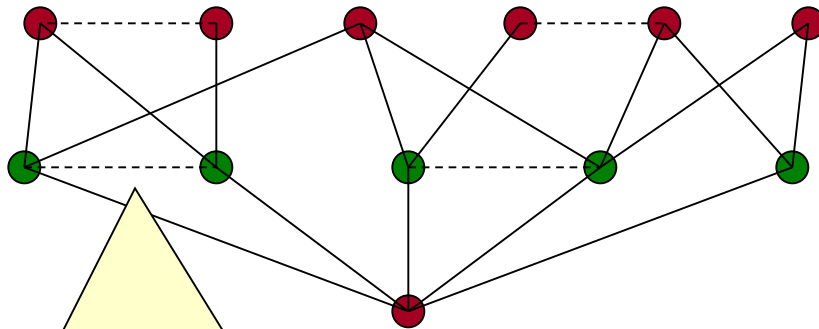
NO

If a graph does not have an odd cycle, then it is bipartite?

Bipartite Graphs

When exactly is a graph **bipartite**?

No such edge because no 5-cycle



No such edge because no triangle

1. The idea is like colouring a tree.
2. Pick a vertex v , colour it **red**.
3. Colour all its neighbour **green**.
4. Colour all neighbours of **green** vertices **red**.
5. Repeat until all vertices are coloured.

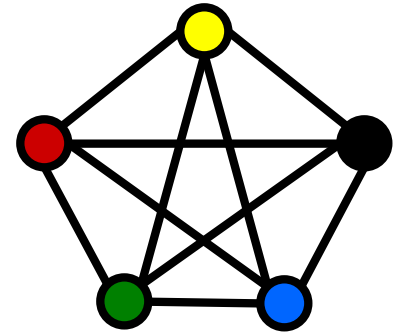
If a graph does not have an odd cycle, then it is bipartite?

Theorem. A graph is bipartite if and only if it has no odd cycle.

Chromatic Number

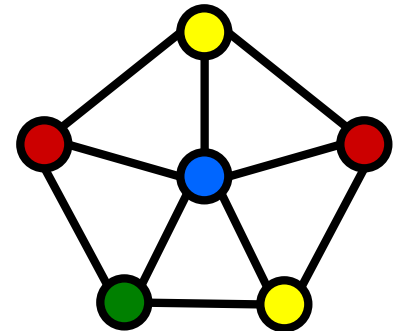
How do we estimate the chromatic number of a graph?

If there is a complete subgraph of size k ,
then we need at least k colours? YES



Is the converse true?

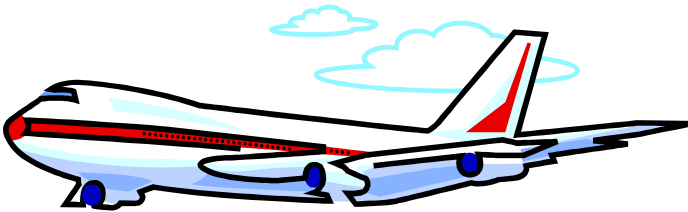
If a graph has no complete subgraph of size 4,
then we can colour it using 4 colours? NO



What graphs are 3-colourable?

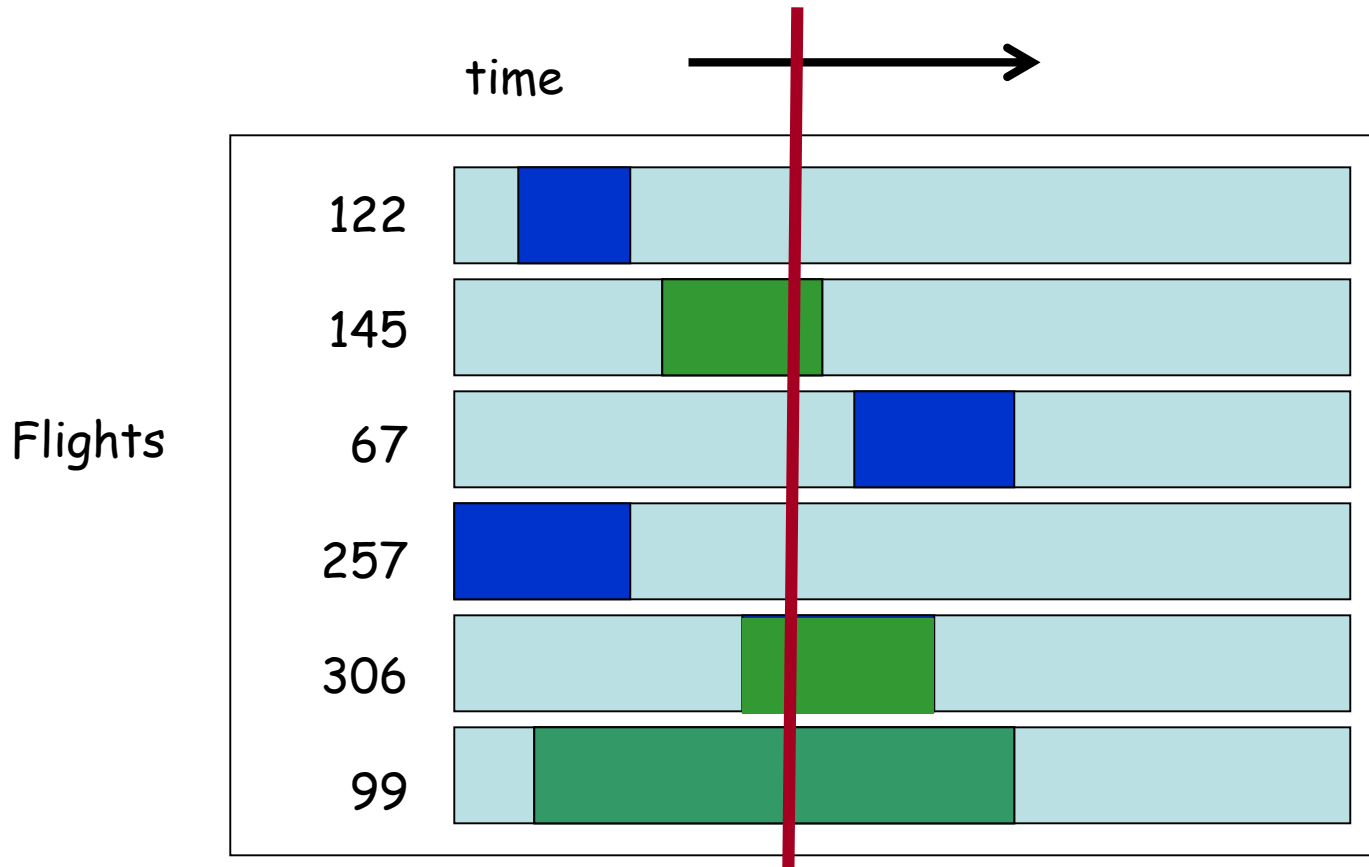
No one knows a good characterization...

Flight Gates



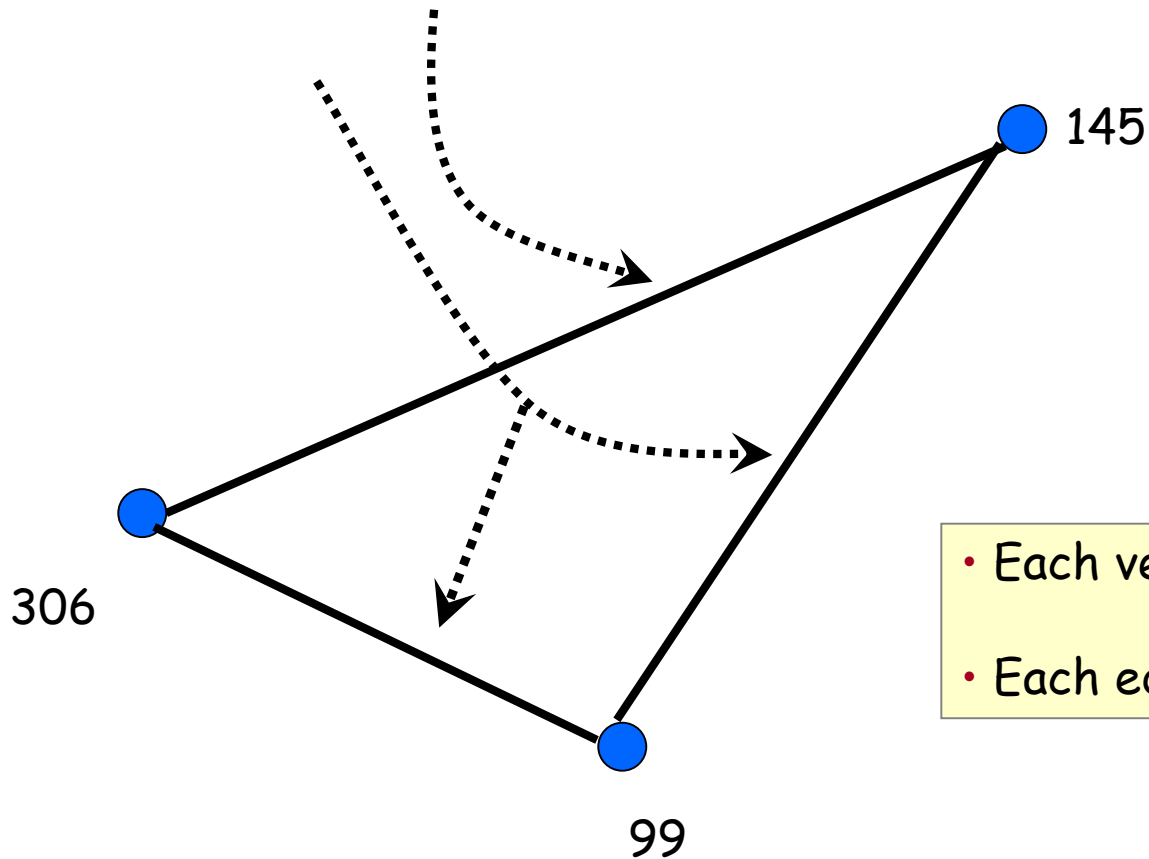
flights need gates, but times overlap.

how many gates needed?



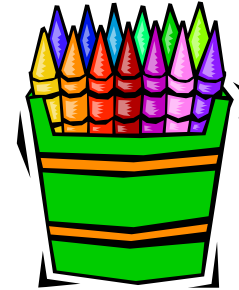
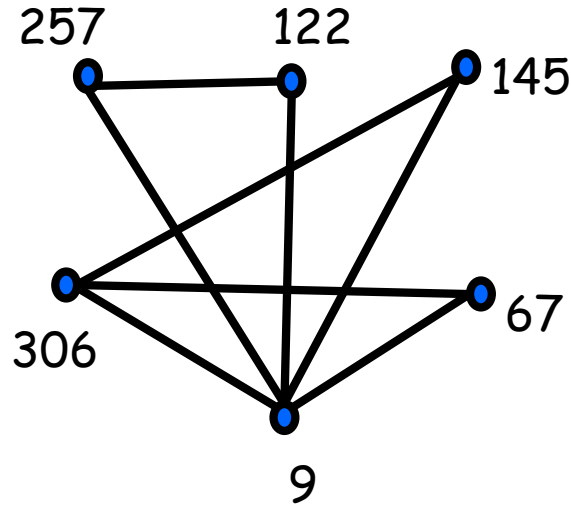
Conflict Graph

Needs gate at same time



- Each vertex represents a flight
- Each edge represents a conflict

Graph Colouring

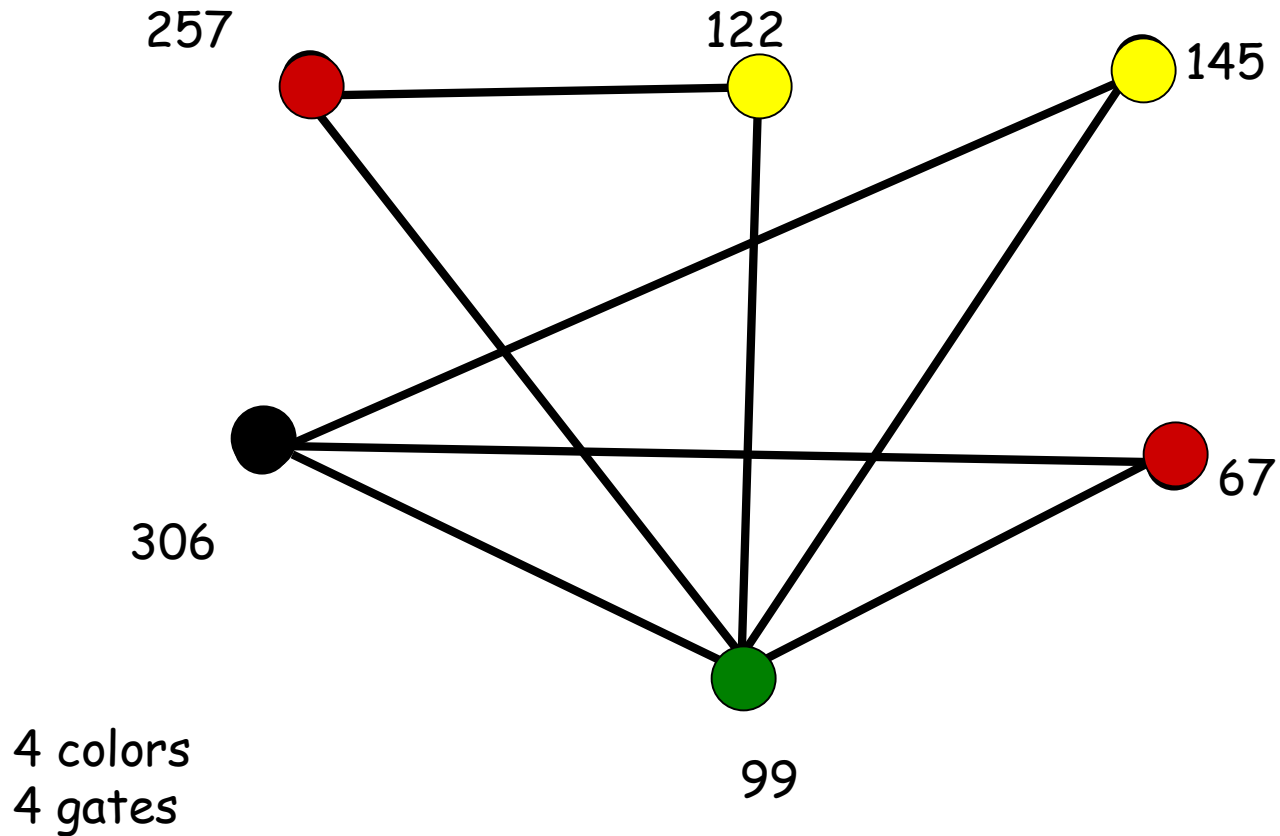


There is a k -colouring in this graph iff the flights can be scheduled using k gates.

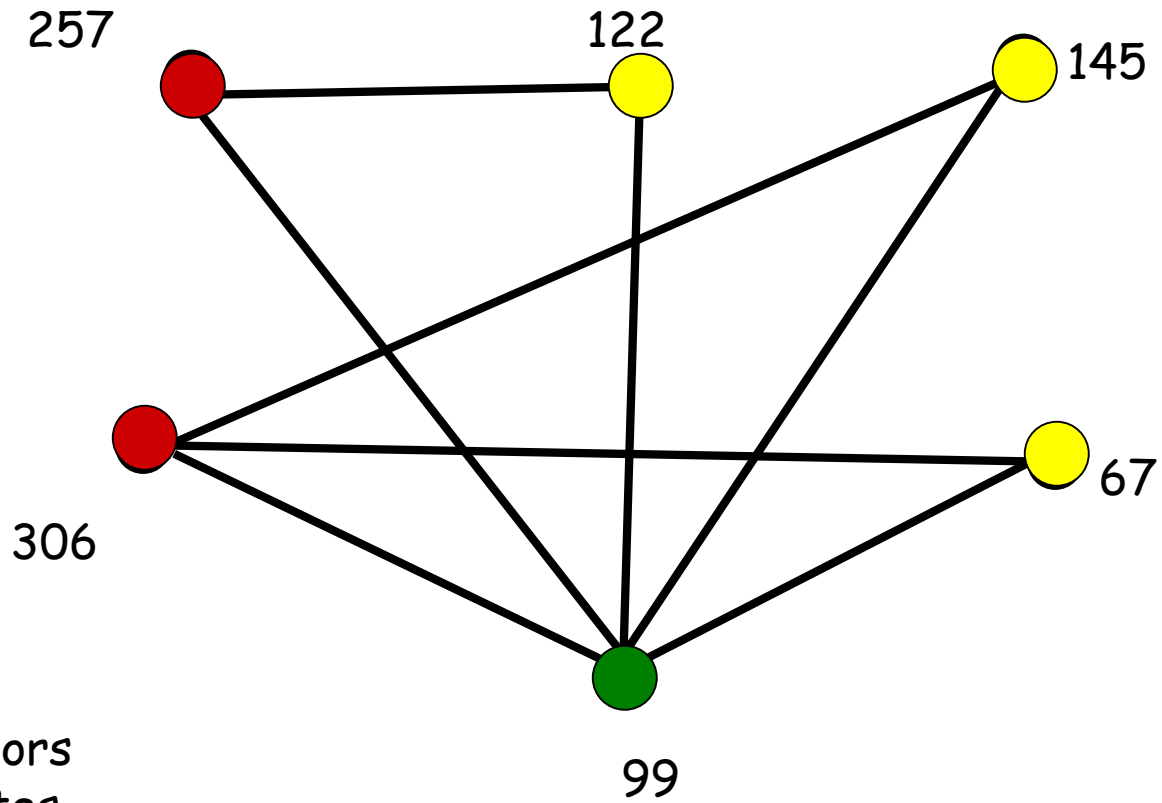
$\Rightarrow$ If there is a schedule, the flights scheduled at the same gate have no conflict, and so we can colour the graph by using one colour for flights in each gate.

$\Leftarrow$ If there is a graph colouring, then the vertices using each colour have no conflict, and so we can schedule the flights having the same colour in one gate.

Colouring the Vertices

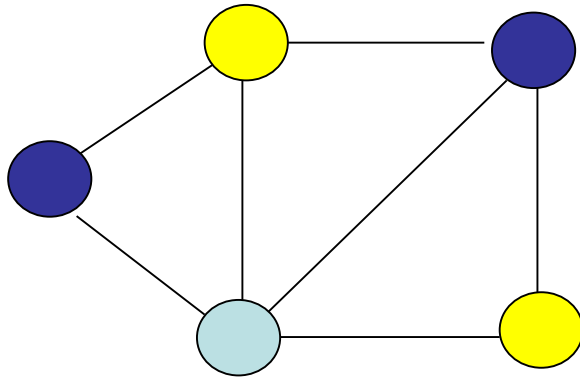


Better Colouring



Coloring Planar Graphs

- **Definition:** A graph is **planar** if it can be drawn in a plane without edge-crossings.



- **The four-color theorem:** For every planar graph, the chromatic number is ≤ 4 .

Was posed as a conjecture in the 1850s. Finally proved in 1976 (Appel and Haken) by the aid of computers.